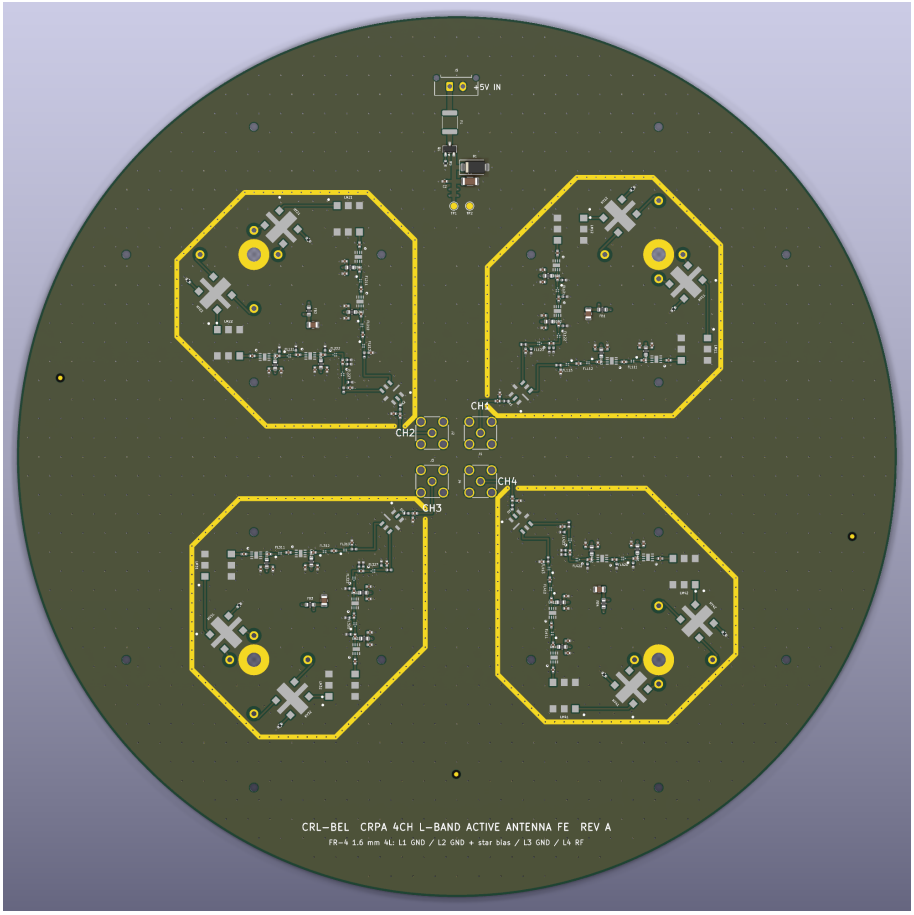
# CRPA 4-Channel L-Band Active Antenna Front End - PCB Design Report

Project 17, Rev A, 2026-11-01. Board: KiCad/CRPA17_4CH_LBAND.kicad_pcb (KiCad 10). Generated by Scripts/make_docs.py; every number below is regenerated from the design data.

## 1. Summary

Four identical RHCP channels on the customer's 200 mm round CRPA board. Antenna module positions (92 mm square pitch, sequential 90 deg rotation), feed-pin positions and the board outline are copied from the customer Gerber (Input/Gerber_4CH_GNSS.zip). Per channel and band: patch 0/90 feeds -> HC125A hybrid -> RLM-63-2W+ limiter -> TQP3M9037 -> SAW -> TQP3M9037 -> SAW -> 100 pF -> SAW, then DXP.02.A combines L1 and L2/L5 onto one SMA output per channel. The chain, values and matching are those of project 16; the limiter (Rev A) and the third SAW per band before the diplexer (Rev B) are new. All LNAs run from one +5 V input (J5). Only the L-band is built (no NavIC S-band), as requested.

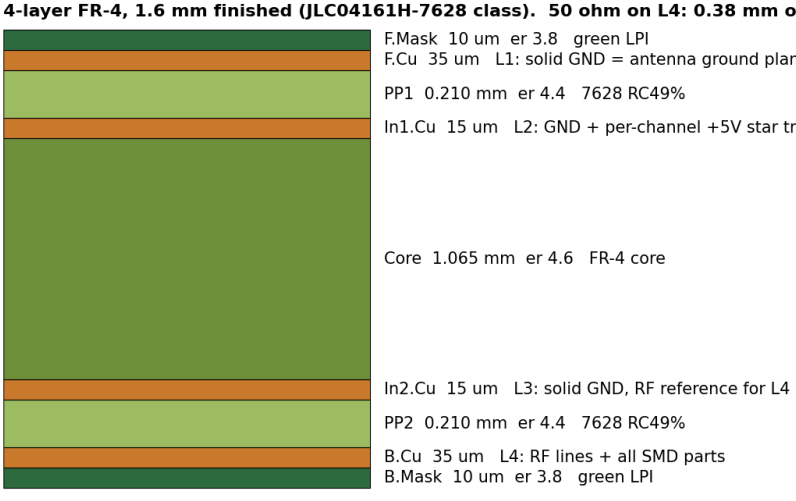
| Item | Result |
|---|---|
| Board | Ø200 mm, 4-layer FR-4 1.6 mm, ENIG, all SMD on the bottom (L4), antennas on top |
| Channels | 4 x (L1 + L2/L5) -> 4 SMA outputs (J1..J4) at the board centre |
| Bias | single +5 V point J5 (Molex Micro-Fit), PTC + reverse-polarity P-FET + TVS; star distribution on L2: one trunk + ferrite + 10 uF per channel |
| Shielding | one soldered shield can per channel (octagonal frame land, GND via fence, RF exit notch), as on the customer product |
| Antenna | Taoglas GPDF6010.A: footprint, drills, clearances and grounded centre screw per its data sheet |
| Parts | 233 placed footprints, 229 fitted (29 BOM lines) |
| DRC (kicad-cli, zones refilled) | 0 violations, 0 unconnected |
| Channel matching | 4 channels are one layout rotated 4x: identical track lengths per net; hybrid feed pairs equal (L1 2.718/2.718 mm, L2 7.082/7.082 mm) |



## 2. Design decisions

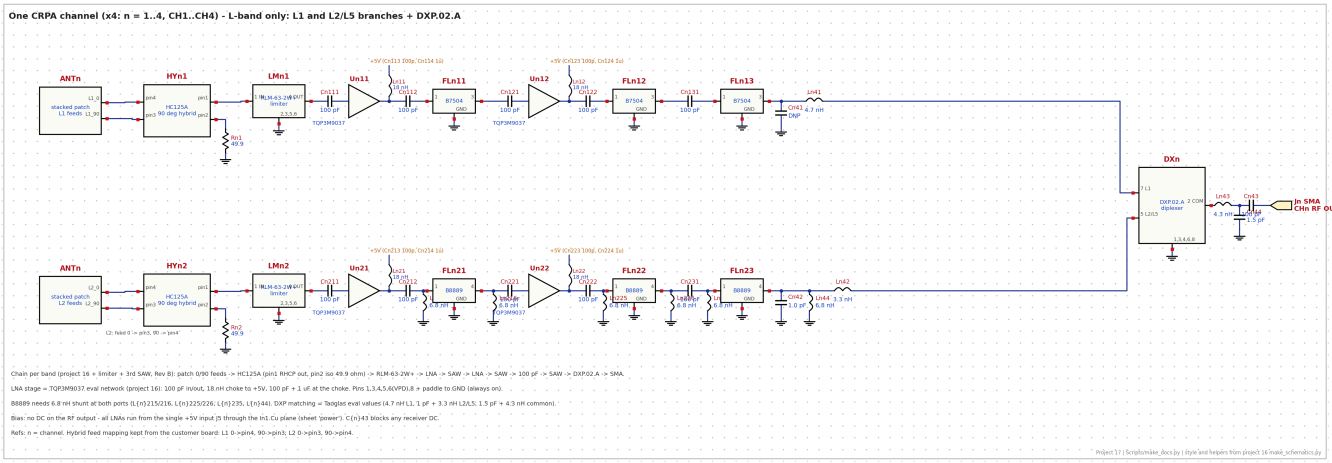
| Topic | Decision | Why |
|---|---|---|
| Substrate | 1.6 mm (not 0.8 mm), 4 layers, FR-4 7628 build | A Ø200 mm board carrying 4 antenna modules needs the stiffness of 1.6 mm. With 4 layers the RF reference (L3) is only 0.21 mm under the traces, so 50 ohm = 0.38 mm, close to the 0402 / DFN / SAW pad widths. On a 2-layer 0.8 mm board 50 ohm would be ~1.5 mm and on 1.6 mm ~3 mm, much wider than every pad (large steps), and the bias would have to cut the ground. |
| Layer use | L1 GND (antenna ground plane), L2 GND + per-channel +5V star traces, L3 GND, L4 RF + parts | Solid ground under the patches as on the original board; bias never crosses an RF reference; every SMD part on one side (one reflow pass). |
| Material | FR-4 (Tg170) instead of RO4350B | Pre-LNA lines are short (see budget): FR-4 adds 0.082 dB/cm, i.e. 0.12-0.16 dB NF for the routed pre-LNA lines. Option: L3-L4 in RO4350B 0.254 mm (hybrid build) saves about half of that. |
| Limiter position | after the hybrid, before LNA1 (one per band per channel, 8 total) | Protects every LNA input; the hybrid (30 W) does not need protection. One limiter per band instead of one per feed halves the count. |
| 3rd SAW (Rev B) | one more SAW per band right before the DXP (B7504 / B8889), 100 pF between SAW 2 and SAW 3 | More out-of-band rejection at the output (customer request). Each B8889 port keeps its own 6.8 nH shunt; the 100 pF gives each shunt its own pad. Costs 0.9-1.5 dB gain, no NF change. |
| Single bias point | dedicated DC connector J5, no DC on any RF output | A bias tee on only one channel would make that channel different (CRPA needs matched channels); 1.1-1.4 A is also too much for a small coax bias inductor. C{n}43 blocks receiver DC. |
| Bias distribution (Rev C) | star: 4 separate 0.8 mm trunks on L2 from the star point after Q1; per channel a ferrite (FB{n}, BLM18KG601) + 10 uF, then 0.4 mm branches to the 4 LNA chokes | Channels share nothing but the star point, so LNA leakage / noise in one channel is isolated from the others by two ferrites; L2 is otherwise solid GND (no +5V plane cavity resonances, third RF ground). |
| Shield cans (Rev C) | octagonal can per channel on a 1.0 mm exposed GND frame land with vias every 2 mm; 2.6 mm notch where the RF output leaves | ~40 dB gain per channel and 4 channels side by side: the cans stop channel-to-channel and external coupling (the customer product uses cans too). Outline: Fab/CRPA17_ShieldCan_outline.csv, Images/shield_can_outline.png. |
| Antenna module (Rev C) | Taoglas GPDF6010.A (from the product photo): D3.2 centre screw hole GROUNDED, 4x D1.8 locating holes at 29 mm, feed pins D0.9 in 1.1 mm holes, D3.5 copper-free around each feed | Data sheet SPE-21-8-111-G sec. 3.1, 4.3, 4.5: the screw is the antenna ground; pins are 2.5 mm long, so they pass the 1.6 mm board. |
| Output connectors | 4 x SMA vertical (Amphenol 132134) at the board centre | Short cable exit under the array, in the patch-free cross between modules, all channels equal. |
| Land patterns | HC125A, TQP3M9037, B7504, B8889, DXP.02.A, antenna: copied from the customer board; RLM-63-2W+: from its data sheet (PL-258) | Customer patterns are proven for their assembly; mirrored back to a top view by Scripts/build_footprints.py. |
| Hybrid pin mapping | kept from the customer board (L1 0->4, 90->3; L2 0->3, 90->4) | Same sense of polarisation as the original board (RHCP at pin 1). |

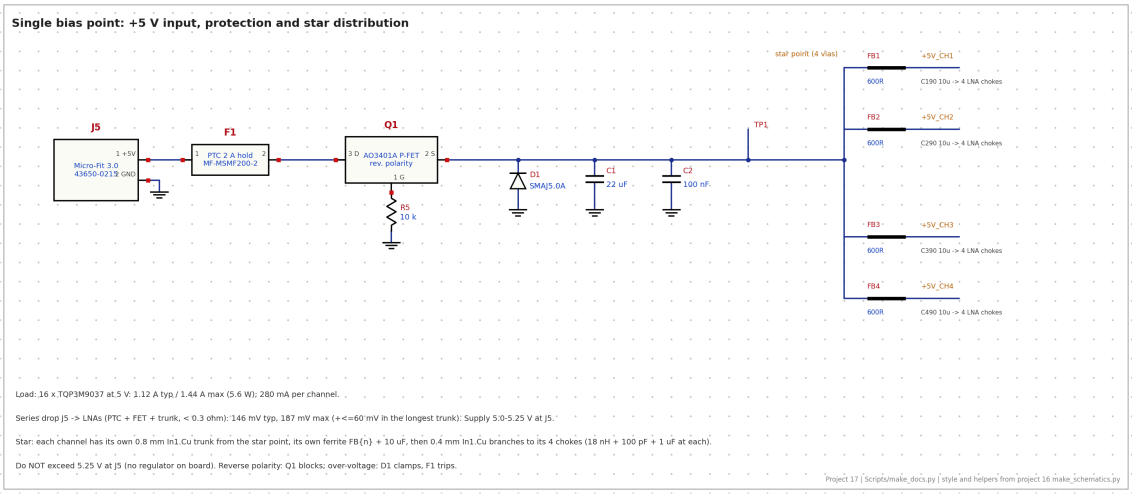
## 3. Stack-up and 50 ohm line



Fab note: impedance control on L4 (50 ohm +/-10 %, 0.38 mm over L3, coupon). Hammerstad-Jensen gives 49.5 ohm for 0.38 mm on 0.2104 mm (er 4.4); the copper pour keeps 0.30 mm from RF copper. Fine-pitch pads are entered with a 0.2 mm x 0.6 mm neck. GND via fences (pitch 2.5 mm, 1.05 mm from the line) along every 50 ohm segment plus a 5 mm stitching grid tie L1/L3/L4.

# 4. Schematic (one channel, x4) and bias



## 5. RF budget (project 16 chain + this board's limiter and lines)

Stage data from project 16 (Extracted_Data/results_cascade_budget.csv, verified there against heavi, circuitRF and Qucs). Added in front of LNA1: the routed line lengths of this board (from the PCB) at 0.082 dB/cm and the RLM-63-2W+ insertion loss (data sheet table, interpolated).

| Case | Pre-LNA line mm | Limiter IL dB | Gain p16 dB | Gain this board dB | NF p16 dB | NF this board dB | dNF dB |
|---|---|---|---|---|---|---|---|
| L1 C/A (1575.42 MHz, 2.046 MHz) | 20.04 | 0.15 | 38.45 | 37.24 | 0.742 | 1.056 | 0.314 |
| L5 (1176.45 MHz, 20.46 MHz) | 15.28 | 0.107 | 42.06 | 40.92 | 0.591 | 0.824 | 0.232 |
| L2C (1227.60 MHz, 2.046 MHz) | 15.28 | 0.113 | 40.09 | 38.4 | 0.622 | 0.861 | 0.239 |

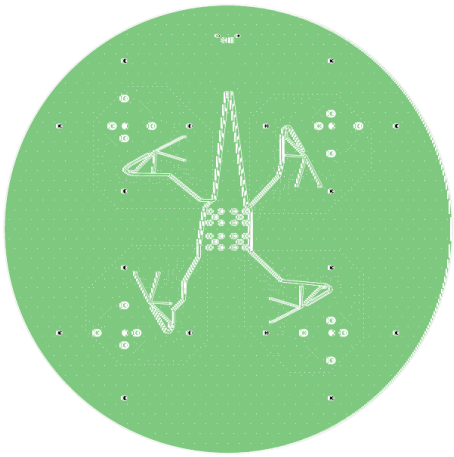
The limiter clamps at about +11.5 dBm (data sheet) - this protects the LNA against damage (strong nearby transmitters, ESD); it does not stop LNA compression, which starts near -1 dBm at the LNA input. Filtering before LNA1 (DME/Iridium) stays an open point from project 16.

## 6. Power

| Quantity | Value |
|---|---|
| LNAs | 16 x TQP3M9037 at +5 V |
| Current | 1.12 A typ, 1.44 A max (280 mA per channel) |
| Dissipation | 5.6 W spread over 16 parts |
| Drop J5 -> LNAs | 146 mV typ / 187 mV max |
| Input range | 5.0 - 5.25 V at J5 (no regulator; TQP3M9037 max 5.25 V recommended) |

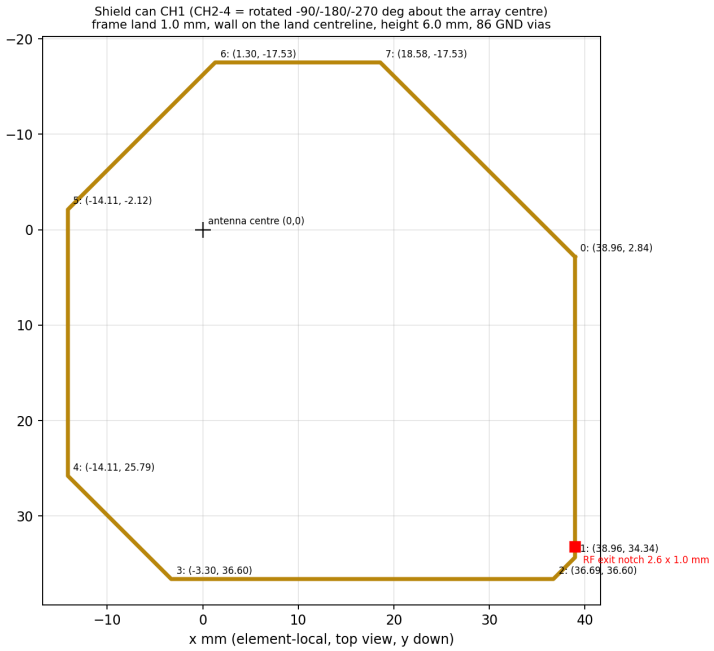
## 6a. Star bias distribution (In1.Cu)

J5 -> F1 -> Q1 -> star point (4 vias). From there each channel has its own 0.8 mm trunk on In1.Cu (CH1 105.3 mm, CH2 106.8 mm, CH3 124.8 mm, CH4 125.1 mm), its own ferrite FB{n} + 10 uF C{n}90, and 0.4 mm In1.Cu branches to its 4 LNA bias vias (18 nH choke + 100 pF + 1 uF at each). The rest of In1.Cu is GND. Routes were found by a grid router that keeps every through-hole clear; CH1's branches are replayed on CH2-4 so all channels stay identical. From the board: 65 bias pads (16 chokes, 4 ferrites), 28 bias vias; DRC 0 unconnected.



## 6b. Shield cans

One can per channel, height 6.0 mm (tallest part: RLM-63-2W+, 4.06 mm max; antenna nut about 2.4 mm). The frame land is 1.0 mm exposed copper (GND) with 86 GND vias per channel; the can wall sits on the land centreline. The RF output leaves through a 2.6 mm x 1.0 mm notch in the wall (red square). Order as a two-piece frame + lid (rework access) or a one-piece drawn can, tin-plated steel 0.2 mm.




## 7. Verification

| Check | Tool | Result |
|---|---|---|
| DRC + zone refill | kicad-cli 10.0.3 pcb drc | 0 violations, 0 unconnected |
| Routing completeness | kicad-happy analyze_pcb.py | complete, 0 unrouted nets |
| Gerber set | kicad-happy analyze_gerbers.py | 4 Cu layers, complete, aligned. GR-005 'outline not closed' = false positive (the outline is a full circle drawn as 2 arcs, as on the customer Gerber). GR-004 paste = expected (no paste on THT/vias). |
| Channel identity | net lengths from the PCB | all 4 channels equal per net |
| Footprint pin maps | customer Gerber X2 pad attributes | HC125A, TQP3M9037, SAWs, DXP: same pin numbers / nets as the customer board; RLM-63-2W+ from the data sheet drawing (pin 1 IN, 4 OUT diagonal) |

## 8. Assumptions and open points

| # | Item | Action |
|---|---|---|
| R1 | RESOLVED (Rev C): Taoglas GPDF6010.A data sheet - pins D0.9 / 2.5 mm long (pass 1.6 mm), holes 1.1 mm, centre screw 3.2 mm grounded, locating holes 1.8 mm, module D60.6 mm | none |
| A3 | Future NavIC S-band patches (product photo: 4 square patches on the axes) are not on this board; the L2 star trunks and the SMA cluster use that area | Decide S-band before Rev D |
| A6 | Antenna nut (bottom) is torqued after reflow; HC125A sits ~6 mm from the screw | Use a thin-wall socket, or fit the antenna before the shield cans |
| A4 | FR-4 er 4.4 / tan d 0.02 for the 7628 prepreg | Fab impedance report; adjust RF_W in crpa_design.py |
| A5 | Land patterns of HC125A, DXP.02.A, B7504, B8889, TQP3M9037 copied from the customer board | Cross-check against vendor drawings (not in the repository) |
| O1 | No pre-LNA band filter (project 16 open point: DME 1150 MHz, Iridium 1621 MHz) | Design decision |
| O2 | Gain ~38-42 dB per channel is high for most CRPA receivers | Add a pad at the output if needed |
| O3 | Vias in the HC125A / TQP3M9037 GND pads | Order resin-plugged + capped vias (IPC-4761 VII) or tent them from the top |

## 9. Bill of materials

| # | Qty | Value | MPN | Mfr | Refs | Fit |
|---|---|---|---|---|---|---|
| 1 | 8 | HC125A | HC125A | Taoglas | HY11 HY12 HY21 HY22 HY31 HY32 HY41 HY42 | SMT bottom |
| 2 | 8 | RLM-63-2W+ | RLM-63-2W+ | Mini-Circuits | LM11 LM12 LM21 LM22 LM31 LM32 LM41 LM42 | SMT bottom |
| 3 | 16 | TQP3M9037 | TQP3M9037 | Qorvo | U111 U112 U121 U122 U211 U212 U221 U222 U311 U312 U321 U322 U411 U412 U421 U422 | SMT bottom |
| 4 | 12 | B7504 | B39162B7504L210 | Qualcomm RF360 | FL111 FL112 FL113 FL211 FL212 FL213 FL311 FL312 FL313 FL411 FL412 FL413 | SMT bottom |
| 5 | 12 | B8889 | B39122B8889P810 | Qualcomm RF360 | FL121 FL122 FL123 FL221 FL222 FL223 FL321 FL322 FL323 FL421 FL422 FL423 | SMT bottom |
| 6 | 4 | DXP.02.A | DXP.02.A | Taoglas | DX1 DX2 DX3 DX4 | SMT bottom |
| 7 | 16 | 18nH | LQW18AN18NG00D | Murata | L111 L112 L121 L122 L211 L212 L221 L222 L311 L312 L321 L322 L411 L412 L421 L422 | SMT bottom |
| 8 | 24 | 6.8nH | LQW15AN6N8G00D | Murata | L144 L244 L344 L444 L1215 L1216 L1225 L1226 L1235 L2215 L2216 L2225 L2226 L2235 L3215 L3216 L3225 L3226 L3235 L4215 L4216 L4225 L4226 L4235 | SMT bottom |
| 9 | 4 | 4.7nH | LQW15AN4N7B00D | Murata | L141 L241 L341 L441 | SMT bottom |
| 10 | 4 | 3.3nH | LQW15AN3N3B00D | Murata | L142 L242 L342 L442 | SMT bottom |
| 11 | 4 | 4.3nH | LQW15AN4N3B00D | Murata | L143 L243 L343 L443 | SMT bottom |
| 12 | 60 | 100pF | GRM1555C1H101JA01D | Murata | C143 C243 C343 C443 C1111 C1112 C1113 C1121 C1122 C1123 C1131 C1211 C1212 C1213 C1221 C1222 C1223 C1231 C2111 C2112 C2113 C2121 C2122 C2123 C2131 C2211 C2212 C2213 C2221 C2222 C2223 C2231 C3111 C3112 C3113 C3121 C3122 C3123 C3131 C3211 C3212 C3213 C3221 C3222 C3223 C3231 C4111 C4112 C4113 C4121 C4122 C4123 C4131 C4211 C4212 C4213 C4221 C4222 C4223 C4231 | SMT bottom |
| 13 | 16 | 1uF | GRM155R61A105KE15D | Murata | C1114 C1124 C1214 C1224 C2114 C2124 C2214 C2224 C3114 C3124 C3214 C3224 C4114 C4124 C4214 C4224 | SMT bottom |
| 14 | 4 | 1.0pF | GJM1555C1H1R0BB01D | Murata | C142 C242 C342 C442 | SMT bottom |
| 15 | 4 | 1.5pF | GJM1555C1H1R5BB01D | Murata | C144 C244 C344 C444 | SMT bottom |
| 16 | 4 | 10uF | GRM21BR61A106KE19L | Murata | C190 C290 C390 C490 | SMT bottom |
| 17 | 1 | 22uF | GRM31CR61A226KE15L | Murata | C1 | SMT bottom |
| 18 | 1 | 100nF | GRM155R71C104KA88D | Murata | C2 | SMT bottom |
| 19 | 8 | 49.9R | ERJ-2RKF49R9X | Panasonic | R11 R12 R21 R22 R31 R32 R41 R42 | SMT bottom |
| 20 | 1 | 10k | ERJ-2RKF1002X | Panasonic | R5 | SMT bottom |
| 21 | 1 | 2A hold | MF-MSMF200-2 | Bourns | F1 | SMT bottom |
| 22 | 1 | AO3401A | AO3401A | Alpha & Omega | Q1 | SMT bottom |
| 23 | 1 | SMAJ5.0A | SMAJ5.0A | Littelfuse | D1 | SMT bottom |
| 24 | 4 | SMA | 132134 | Amphenol RF | J1 J2 J3 J4 | Hand solder (THT) |

| # | Qty | Value | MPN | Mfr | Refs | Fit |
|---|---|---|---|---|---|---|
| 25 | 1 | +5V IN | 43650-0215 | Molex | J5 | Hand solder (THT) |
| 26 | 2 | TP | - | - | TP1 TP2 | SMT bottom |
| 27 | 4 | DNP | - | - | C141 C241 C341 C441 | DNP |
| 28 | 4 | L1/L2 stacked patch | existing module | customer | ANT1 ANT2 ANT3 ANT4 | Customer supplied, hand fit (top) |
| 29 | 4 | 600R@100MHz | BLM18KG601SN1D | Murata | FB1 FB2 FB3 FB4 | SMT bottom |

## 10. Files

| Path | Content |
|---|---|
| Fab/CRPA17_4CH_LBAND_Gerbers.zip | Gerber X2 + Excellon (PTH/NPTH) + drill map + job file |
| Fab/CRPA17_4CH_LBAND_BOM.csv / .xlsx | BOM |
| Fab/CRPA17_4CH_LBAND_CPL_bottom.csv | pick-and-place (bottom side, mm, deg) |
| Fab/CRPA17_4CH_LBAND.d356 | IPC-D-356 netlist (bare-board test) |
| KiCad/ | board, project, DRC rules, footprint library CRPA17.pretty |
| Docs/ | this report, layer PDF, netlist, RF budget CSVs |
| Scripts/ | build_footprints.py, build_pcb.py, run_drc.sh, export_fab.sh, make_docs.py, build_all.sh |